

Dynamic Model Implementation

Chapter 6 - 2SLS, Causality Testing, and Dynamic Modeling

[Dynamic Model Implementation | Springer Nature Link](#)

Objectives

- Understand and implement Two-Stage Least Squares (2SLS)
- Learn Granger causality testing
- Implement Autoregressive (AR) models
- Apply Koyck transformation
- Master ARDL model implementation
- Understand ARMA and ARIMA modeling
- Implement rolling forecasts

Chapter 6 is even MORE demo-focused than Chapter 5

1. Procedural/Sequential Nature

- Chapter 6 follows a strict workflow:
Stationarity Check → Transform → Model Selection → Fit → Diagnose → Forecast
- Each step must be executed to see the next step work.

2. Complex Mathematical Transformations

- 2SLS: Two-stage process with predicted values
- Koyck: Geometric decay demonstration
- ARIMA: Differencing, parameter selection
- Rolling Forecast: Iterative updating
- These are impossible to understand from slides alone.

3. Forecasting is Visual

- Forecast plots show prediction vs reality
- Confidence intervals are visual
- Rolling forecast accuracy is visual
- PPT cannot capture dynamic forecast evolution

4. Model Selection is Interactive

Grid Search → Try combinations → Compare AIC → Select Best

Students need to see:

- Why some models fail
- How AIC changes
- Trade-offs in complexity

5. Diagnostic Tests are Shown

- Ljung-Box test results
- Residual plots
- ACF/PACF analysis

These require live demonstration to interpret.

Two-Stage Least Squares (2SLS)

Problem: Endogeneity

- Explanatory variable correlated with error term
- OLS estimates become **biased and inconsistent**

Sources of Endogeneity:

- Reverse Causality: Y causes X
- Omitted Variables: Missing variables affect both
- Measurement Error: Inaccurate measurement

Solution: Instrumental Variables (IV)

- Correlated with endogenous variable
- Uncorrelated with error term
- Exogenous (not directly affecting Y)

2SLS Process

Stage 1: Predict Endogenous Variable

$$InterestRate_t = \pi_0 + \pi_1 \times MoneySupply_{t-1} + v_t$$

- Regress endogenous variable on instrument(s)
- Obtain predicted values $\widehat{InterestRate}_t$

Stage 2: Estimate Original Model

$$GDP_t = \alpha + \beta \times \widehat{InterestRate}_t + \epsilon_t$$

- Replace endogenous variable with predicted values
- Run OLS regression

Key Requirement: F-statistic > 10 in Stage 1 (strong instrument)

2SLS with Simultaneous Equations

Problem: Inflation and Stock Returns

Inflation Equation:

$$Inflation_t = \alpha_0 + \alpha_1 CL_ExcessReturns_t + \alpha_2 ConsumerCredit_t + \varepsilon_{1t}$$

CL_ExcessReturns Equation:

$$CL_ExcessReturns_t = \beta_0 + \beta_1 SP_ExcessReturns_t + \beta_2 Inflation_t + \beta_3 ConsumerCredit_t + \varepsilon_{2t}$$

Endogeneity Issues:

- CL_ExcessReturns endogenous in inflation equation
- Inflation endogenous in CL_ExcessReturns equation
- Use SP_ExcessReturns and ConsumerCredit as instruments

Granger Causality

Concept:

- Tests if past values of X help predict future Y
- Not true causation, but predictive causality

Hypotheses:

- H_0 : X does not Granger-cause Y
- H_1 : X does Granger-cause Y

Decision Rule:

- $p\text{-value} < 0.05 \rightarrow \text{Reject } H_0 \text{ (X Granger-causes Y)}$
- $p\text{-value} \geq 0.05 \rightarrow \text{Fail to reject } H_0$

Koyck Transformation

Koyck Transformation

Purpose:

- Simplify Infinite Distributed Lag (IDL) models
- Assume geometric decay of lagged effects

Original IDL:

$$Y_t = \beta_0 + \beta X_t + \beta\lambda X_{t-1} + \beta\lambda^2 X_{t-2} + \dots + \varepsilon_t$$

After Koyck Transformation:

$$Y_t = \alpha_0 + \alpha_1 Y_{t-1} + \beta X_t + \varepsilon_t$$

Parameters:

- $\alpha_0 = \beta_0(1 - \lambda)$
- $\alpha_1 = \lambda$ (persistence parameter)

ARMA Models

Components:

- AR (Autoregressive): Y_t depends on past values
- MA (Moving Average): Y_t depends on past errors

Model:

$$Y_t = \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q} + \varepsilon_t$$

Key Challenge:

- Lagged errors are unobservable
- Cannot use OLS
- Use Maximum Likelihood Estimation (MLE)

Box-Jenkins Method:

- Identification: ACF/PACF plots
- Estimation: MLE
- Diagnostic Checking: Residual analysis

Questions & Discussion

Review Questions:

- When is 2SLS preferred over OLS?
- What does Granger causality actually test?
- Why is stationarity important for AR models?
- How does the Koyck transformation simplify IDL models?
- What is the difference between ARMA and ARIMA?

Discussion Points:

- Choosing between static and rolling forecasts
- Challenges with dynamic model identification
- Interpretability vs accuracy trade-offs